

Yashwanth Kasanneni

New York, United States | +1 347-325-6379 | kasanneni29@gmail.com | [GitHub](#) | [LinkedIn](#)

EDUCATION

NYU Tandon School of Engineering

Master's in Computer Science

Coursework: Analysis of Algorithms, Machine Learning, Distributed Systems, Principles of Database Systems

Expected May 2027

New York, United States

WORK EXPERIENCE

Software Developer Intern

Sept 2023 - Dec 2023

Enabled Analytics

On-Site

- Developed a custom Salesforce-based order management system integrated with WhatsApp, allowing supermarket customers to browse items, place orders, and receive automated responses through a chat-based workflow.
- Built Apex, SOQL/SOSL, and Salesforce Flow automation to process WhatsApp orders in real time, validate item availability, update inventory, prevent duplicate orders, and sync order data across Salesforce records.
- Created Salesforce Lightning interfaces for supermarket staff to monitor WhatsApp orders, update fulfillment status, manage inventory, and access customer/order history from a centralized dashboard, owning the feature end-to-end from design through testing and production deployment while debugging workflow-execution issues across order placement and inventory sync.

PROJECTS

BeforeYouSign - Real-Time Conversational AI Agent | Python, FastAPI, React, TypeScript, WebSockets, Gemini Live API, Google Cloud Run

- Built a real-time conversational AI agent that handles live phone and chat sessions end to end, bridging bidirectional audio over WebSockets to a Gemini Live model and returning low-latency spoken and text responses through a React and TypeScript frontend.
- Grounded the agent in live external records using LLM tool calling, running concurrent asynchronous lookups and deterministic scoring while preserving multi-turn session context, so automated answers stayed accurate instead of relying on model knowledge alone.

Recall - AI-Native Multimodal Search and Knowledge Platform | TypeScript, Next.js, GraphQL, Python, FastAPI, AWS, OpenSearch, PostgreSQL

- Built a cloud-native, user-facing product that answers natural-language queries over a personal file library, backed by an event-driven AWS pipeline (S3, SQS, ECS workers) that asynchronously indexes content and a hybrid lexical and semantic retrieval engine on OpenSearch.
- Shipped a Next.js and TypeScript interface served through a GraphQL API, owning the feature from ingestion pipeline to frontend, with idempotent processing, retries and dead-letter queues for fault tolerance, and strict per-user data isolation across storage and search.

PolicyDiff - AI-Powered Change Detection and Automation Pipeline | Python, TypeScript, Next.js, FastAPI, Gemini, ClickHouse, Datadog

- Co-built a microservice pipeline that leverages an LLM to classify document changes into structured, machine-readable events, routing low-confidence outputs to human review rather than auto-publishing so the automated decisions stayed reliable in production.
- Instrumented every model call with LLM observability to trace prompts, outputs, and token usage, and surfaced results through a Next.js and TypeScript dashboard over an event-driven store, keeping AI confined to the semantic step while deterministic logic handled detection and calculation.

CourtIQ - Real-Time Analytics Platform (Lambda Architecture) | Python, TypeScript, React, Kafka, Spark Structured Streaming, FastAPI, PostgreSQL

- Built a real-time analytics platform on a Lambda architecture, pairing a Kafka and Spark Structured Streaming speed layer for live metrics with a batch layer, so one event stream served both instant and historical views behind operational dashboards.
- Served the merged views through a FastAPI backend and shipped a React and TypeScript frontend with interactive dashboards, running the full multi-service stack in Docker and completing an architectural gap analysis of its failure modes and scaling limits.

RESEARCH AND PUBLICATIONS

[Effective Analysis of Machine and Deep Learning Methods for Diagnosing Mental Health Using Social Media Conversations](#)

IEEE Transactions on Computational Social Systems

- Analyzed 100K+ tweets and developed NLP pipelines using BERT, RoBERTa, DistilBERT, XLNet, and classical/deep-learning models for behavioral risk-pattern detection.
- Evaluated model accuracy and sentiment features to build predictive systems that identified psychological-condition signals with 97% accuracy.

SKILLS

Languages and Frameworks: Python, TypeScript/JavaScript, SQL, React, Next.js, Node.js, FastAPI, GraphQL, data structures and algorithms

Full-Stack and Product Engineering: React, Next.js, TypeScript, REST and GraphQL APIs, real-time UIs, WebSockets, end-to-end feature ownership

AI and Conversational Systems: LLM integration (Gemini), tool calling, agentic AI, human-in-the-loop review, real-time voice and chat agents, NLP, LLM observability

Cloud, Data and Real-Time: AWS (S3, SQS, ECS, RDS, OpenSearch), Docker, Kubernetes, Kafka, Spark Structured Streaming, PostgreSQL, ClickHouse, Redis, event-driven pipelines

Certifications: Machine Learning Specialization by Andrew Ng, Google Cloud Digital Leader